Automated Milk Feeders and the Reconfiguration of Calf Care

A Qualitative Case Study of Socio-Technical Entanglements on Swedish Dairy Farms

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This thesis explores how automated milk feeders (AMFs) for calves participate in reconfiguring care practices on Swedish dairy farms. Drawing on perspectives from Animal-Computer Interaction, more-than-human perspectives and Actor-Network Theory (ANT), calf care is approached as emerging through relations between farmers, calves and the technological system. The study was conducted as a qualitative case study based on semi-structured interviews with farm workers, veterinarians, advisors and researchers. Through thematic analysis, four themes were identified: Reorganizing Care Practices, The Socio-Technical Stockperson's Eye, Distributing Responsibility and Control and Calves as Active Participants. The findings show that AMFs do not simply automate feeding, but reconfigure how attentiveness and responsibility are enacted and organized. In particular, the thesis identifies the emergence of a socio-technical stockperson's eye, where embodied and experiential knowledge becomes intertwined with digital monitoring and data interpretation. Rather than replacing human attentiveness, technology participates in reshaping and extending it. At the same time, the findings demonstrate that the effects of AMFs are highly dependent on situated farm contexts. The thesis therefore argues that livestock technologies cannot be understood or designed as isolated tools, but must be approached as embedded within broader socio-technical systems of care. By applying more-than-human and ANT perspectives to agricultural HCI, the thesis contributes to discussions on livestock technology and One Welfare.

CCS CONCEPTS • **Human-centered computing–Human computer interaction (HCI)–Empirical studies in HCI** • Human-centered computing–Human computer interaction (HCI)–HCI theory, concepts and models • Applied computing–Computers in other domains–Agriculture

Additional Keywords and Phrases: Automated milk feeders, Animal-Computer Interaction, Actor-Network Theory

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1 INTRODUCTION

Automated milk feeders (AMFs) for calves are increasingly being adopted in dairy farming to reduce labor demands and support calf health through more natural feeding patterns and improved individual monitoring. By distributing milk in smaller portions throughout the day and tracking intake data, AMFs are often framed as technologies that benefit both productivity and animal welfare [39, 43, 44, 46].

However, AMFs are also transforming calf care by introducing new routines of monitoring and data interpretation on farms [17, 18, 44]. When this process becomes technologically automated, farmers may increasingly come to know their animals through data representations rather than through physical presence. This shift may reshape how care is enacted [5, 24]. Despite this, AMFs and other agricultural technology have mostly been studied through a productivity-oriented lens, emphasizing efficiency, cost reduction and risk management [1, 28, 36]. Other research has emphasized that agricultural

technologies cannot be understood independently of the situated practices in which they are used. One argues that these technologies only become meaningful through farmers' experiential and situated knowledge and should therefore be understood as embedded within broader socio-technical systems rather than as isolated technical tools [30]. Another calls for examining agricultural technologies from a care perspective, where care is viewed not as something performed separately from technology, but as something emerging through situated relations between humans, animals and technical systems [29].

Importantly, there is a central actor in this equation that is often underexamined: the animals themselves. Within Human-Computer Interaction (HCI), the expansion toward Animal-Computer Interaction (ACI) has emphasized that animals also interact with and are affected by digital systems. However, limited research has applied HCI or ACI perspectives to agricultural technologies. Calves, in particular, have received far less attention than adult dairy cows [32, 36, 49]. Moreover, despite claims of being animal-centered, much ACI research remains mostly oriented toward human goals [32, 36, 49].

The concept of One Welfare seeks to address this gap by connecting animal welfare, human well-being and environmental sustainability [40]. From a One Welfare perspective, these elements cannot be separated. Farmer well-being affects animal care, animal welfare influences farmer satisfaction and economic outcomes and both are shaped by infrastructures, technologies and ecological conditions [3]. This perspective highlights the importance of recognizing the broader network of actors involved.

From the perspective of Actor-Network Theory (ANT), human and animal actions and intentions are made possible and shaped by material actors, including technologies, which ANT understands as possessing a form of agency [15, 25, 42]. ANT was used as a theoretical lens in this thesis and is particularly suited to the study of AMFs since these systems actively participate in shaping how farmers attend to calves and how calves engage with feeding. ANT allows us to see how AMFs mediate interactions between calves and farmers, and how these mediations transform care. Through this perspective, AMFs are not neutral tools but active participants in structuring care practices and may, in the long run, contribute to welfare outcomes. Aspling and Juhlin conducted a study arguing that ANT provides a useful contribution to ACI, concluding that technologies shape the possibilities for action and that outcomes are produced through the interplay of humans, animals and technologies [2].

1.1 Aim and Research Questions

This thesis aims to explore how AMFs participate in reconfiguring calf care practices on dairy farms by approaching calf care as distributed across humans, animals and technologies, and the AMF system as actively shaping how farmers attend to calves. In doing so, it also seeks to examine how calves themselves respond to and participate in care. The thesis will do this by drawing on ANT and investigating how care practices, in the form of attentiveness, responsibility and calf behaviors, emerge through these interconnected networks on dairy farms. Particular attention will be given to how farmers' embodied and experiential ability to interpret and respond to signs of animal health and illness is reconfigured. Lastly, this thesis aims to contribute to an understanding of livestock technology not as neutral tools, but as active participants in socio-technical systems of livestock care. To better understand what AMFs introduce into calf care practices, this thesis also considers manual feeding as a point of comparison. Examining manual feeding practices makes it possible to identify how care is organized without automation and thereby highlight what is reconfigured when AMFs are introduced. While the primary focus remains on the technology, this comparative perspective helps situate AMFs within existing care practices rather than treating them in isolation. To address this aim, the following research questions have been formulated:

RQ1. How do farmers, calves and automated milk feeders participate in reconfiguring calf care on dairy farms?

RQ2. How are attentiveness and responsibility reshaped within these socio-technical systems compared to manual feeding systems?

In this thesis, calf care is understood as the practices through which calves' health and daily needs are attended to on dairy farms. Attentiveness refers to the engagement and ability to notice, interpret and respond to behavioral and physical signs in calves. Responsibility is understood as the obligation to ensure that calves receive appropriate care and to maintain oversight and control of the care process.

2 RELATED WORK

2.1 The Automation of Livestock

2.1.1 Automated Livestock Technology

Earlier research argues that the implementation of precision livestock farming (PLF) technologies effectively transforms livestock care [5, 11, 17, 24, 31, 38]. PLF refers to the use of technologies that enable real-time monitoring and automated decision-making in livestock farming, such as automated feeding and milking systems, sensors, cameras as well as AI and machine learning [38].

There are concerns regarding PLF's effects on the degree of closeness or distance between farmers and their animals. Some argue that PLF has reshaped the farmer's role from that of a caregiver to more of a system supervisor, as dependence on technology grows. Rather than being directly engaged in day-to-day care, farmers increasingly act as managers of technical systems, which may weaken traditional stockmanship skills and embodied, experiential knowledge [5, 24, 38]. At the same time, physical contact between humans and animals tends to decrease, and opportunities for positive human-animal relationships are reduced, making genuinely attentive care harder to achieve [5, 38]. There is also a risk that animals become objectified in the eyes of farmers, as they are increasingly represented as data points and quantifiable metrics. Such representations can oversimplify animals' complex emotional and behavioral needs, and algorithms may misinterpret behaviors [5, 38]. As farming practices become increasingly quantified, the processes of food production also risk becoming less visible and harder for consumers to relate to [5].

At the same time, one study highlights more positive relational effects of PLF technologies, suggesting that they can open up for new forms of interaction between farmers and animals, particularly during transitional and familiarization phases. The authors also emphasize that whether such technologies increase or reduce human-animal distance depends on the farmer's profile and how the equipment is used [24].

2.1.2 Automated Milk Feeders

Cows are as ruminants, meaning animals that digest food through a specialized stomach system adapted for breaking down grass and hay. Calves are born as non-functional ruminants and therefore initially depend on milk feeding before gradually transitioning to solid feed as the rumen, the largest of a cow's four stomach compartments, develops [23]. In natural settings, calves consume milk in small, frequent meals. AMFs allow feeding practices that more closely resemble these natural patterns compared to traditional manual feeding systems, which often distribute larger portions at fixed times. Today, many dairy farms still use manual feeding systems in which calves are typically fed milk twice per day [45]. These systems often involve buckets with artificial teats designed to simulate natural suckling, sometimes with the help of mobile milk taxis used to mix, transport and dispense milk [33].

AMFs were developed to support more frequent and individualized feeding while also reducing labor demands and enabling closer monitoring of calves. With this technology, calves decide when and how much to drink, within certain constraints, and are therefore fed smaller amounts throughout the day rather than all at once, which has been linked to improved health and growth [39, 43, 44, 46]. Historically, calves were often fed limited amounts of milk to reduce costs and to encourage earlier consumption of solid feed in order to stimulate rumen development [45]. AMFs allow larger milk allowances and more gradual weaning without substantially increasing labor demands, which have been associated with reduced stress for calves and continued growth [43, 45, 46].

The system consists of different feeding stations with teats for the calves to drink from. Each calf wears an RFID ear tag that identifies it at the feeding station, allowing the machine to prepare a customized milk recipe. The system is individualized for each calf using a "fuzzy algorithm" that enables flexible decision-making based on input parameters such as age, weight and behavior to determine the appropriate recipe. The system heats water, adds the correct amount of milk powder, and includes vitamins or medicine if needed, then mixes everything automatically. The calf can then drink from the teat, and the system measures how much milk it consumes. After feeding, the machine cleans itself to prevent diseases from spreading between calves. The machine also has an alarm system that provides warnings if something fails, helping to keep the calves safe. Farmers can control everything through an interface where they can view calf data and manage recipes, schedules, alarms and RFID tags [20]. Figures 1–2 illustrate different components of an AMF system, including the feeding station used by calves and the mixing unit responsible for preparing individualized milk portions.

Although AMFs reduce physical labor, they also introduce new demands for farmers and require more initial training, such as developing skills in data interpretation and monitoring, as well as training calves to use the system [18, 44]. This is particularly relevant since some calves struggle to adapt to automated systems [26]. AMFs may also reduce human-animal interactions, meaning farmers may become less experienced in recognizing physical and behavioral signs in calves [43, 50].

Additionally, these systems typically involve group housing. As a result, social dynamics may emerge in which calves compete for access to the feeder while also learning feeding behaviors from one another. Reducing group size or increasing feeder availability are possible strategies for limiting competition [22]. AMFs can also monitor each individual calf's milk intake and visits to the feeding station, helping farmers detect unusual behaviors and identify disease [35, 39, 43]. In manual feeding systems, individual changes among group-housed calves may be harder to notice [35]. However, AMF systems may also increase spreading of illness if not managed correctly [18].



Figure 1: Feeding station: (a) calf pen equipped with a feeding station and (b) close-up of feeding station with teat for drinking (AI-illustrated from original photographs).



Figure 2: Mixing unit where individualized milk portions are prepared (AI-illustrated from an original photograph).

2.2 Animal-Computer Interaction

HCI has expanded to include animals through ACI, as they also interact with technology. This includes the idea of Human-Animal Interactions, which are important for both humans and animals, and can be experienced as either positive or negative [3, 50]. Machine-Animal Interactions refer to animals' interactions with technological systems that, in farming contexts, may replace or support traditional human labor [50].

Many researchers call for more animal-centric approaches to design. They argue that, despite claims of being animal-centered, the field of ACI has often remained anthropocentric by designing primarily for human purposes, treating animals as passive users rather than active participants, or by failing to acknowledge the inherent inequality between humans and animals [8, 32, 48–50]. Some argue that we need to think more about animals' experiences, support their activities and foster positive Human-Animal Interactions [32, 48, 50]. This also involves recognizing animals as individuals and giving them choices and greater control over their environments [48, 50]. Since animals cannot provide verbal explanations, attention must instead be paid to behaviors, preferences and engagement [8, 19, 47, 48].

At the same time, researchers recognize the ethical difficulties involved in designing with animals in mind, with some arguing that animals will always be understood through a human perspective and that humans ultimately remain the designers [19, 32, 49]. Human meaning is often projected onto other species, a process referred to as anthropomorphism [8, 32, 48, 49]. Rather than ignoring this, it is important to continue recognizing and reflecting on the unequal relationship between humans and animals [32, 49]. Others, however, argue that true co-design with animals is possible [47], or that that we can aim for equal participation in the future [8, 19].

2.3 More-Than-Human Perspectives and Design

More-than-human perspectives are increasingly used in HCI, criticizing the field for being anthropocentric and for privileging human needs and agency. They are grounded in traditions such as posthumanism, which rejects placing the human at the center of all systems, as well as new materialism, which understands both humans and nonhumans as entities with agency that emerge together through their relations [12, 16, 37, 41]. This development has also been a response to the growing need to address climate and ecological concerns, as well as the complex socio-technical world we live in today [12, 16, 27, 34, 37, 41].

More-than-human perspectives recognize that humans are entangled with nonhumans and technologies, and that they all exist within systems that are messy and constantly evolving [12, 16, 37]. In this view, humans are not the only ones with agency. Animals, technologies, plants, materials, ecosystems and infrastructures are seen as active participants, and human action cannot be understood as separate from them [34, 37, 41]. It is important to note that decentering the human does not mean that we are unimportant, but rather that we need to reimagine how we understand and position the human [37].

More-than-human design can be understood as the practical application of these perspectives. It involves formulating problems and asking questions not only for human purposes or from human viewpoints, but also exploring nonhuman ways of experiencing and knowing [34, 37]. Within more-than-human perspectives, design is not seen as neutral, but can both reinforce or challenge anthropocentric assumptions [12, 16]. Design can be approached through research through design or speculative design to explore and generate knowledge about situations in which nonhumans are treated as stakeholders. Ethnographic approaches are also used to study real entanglements involving humans and nonhumans in action [16, 37].

More-than-human design further calls for more participation of nonhuman stakeholders in design processes [16, 34]. For example, Moreira's concept of representational participation refers to representing nonhumans through caretakers or data, while material representation looks at what nonhumans actually do within a system and how they shape it. The latter

considers how animals communicate needs through behavior, how technologies communicate through functionality or breakdowns and how other parts of the system respond, such as farmers adjusting their care practices [34].

2.4 Actor-Network Theory as Theoretical Lens

ANT was developed by Bruno Latour and aligns with more-than-human thinking, as it views material things as actors with agency and does not assume that humans are fundamentally special. These actors can include technologies or buildings, but also non-material entities like laws and societal norms [15, 25]. Although animals are not always explicitly included, some research has explored or mentioned them [2, 42].

Within ANT, agency is not about intention or morality, but about what makes a difference in the course of action. Actors are understood as shaping action through their connections with other actors [15, 25, 42]. Law illustrates this by arguing that human activities do not occur solely within our bodies, but through connections that extend beyond us. For instance, almost all human interactions are enabled and shaped by objects, such as communicating through a phone [25]. Sayes explains this by framing nonhumans as mediators that transform relations, but the underlying idea is the same [42].

The purpose of ANT is to analyze complex situations in which multiple diverse actors have influences, revealing new insights [27]. Law argues that when networks of actors function smoothly, they appear as one stable entity, and the underlying complexity disappears from view. Reality is *enacted* rather than observed. Yet this stability must constantly be reproduced and is never guaranteed; networks can always break down. ANT also allows us to trace these configurations and make new things visible [25].

It is important to note that ANT has limitations. Its claim that agency is distributed across actors in a way that makes them equally important in producing social effects is both ontologically and ethically controversial [15, 25, 42]. Consequently, some argue that ANT should be understood as an analytical tool meant to prevent the exclusion of nonhuman actors, rather than as a moral claim [25, 42]. Others suggest that ANT's claims make more sense methodologically and strategically, rather than theoretically [15, 42]

Aspling and Juhlin suggest a new framework for studying ACI using ANT that accounts for the multiple actors involved, such as caretakers, designers, animals, devices, tools, infrastructures and the environment, rather than a direct one-to-one interaction between animal and system. They use the idea of machinations to show that neither the animal nor the technology acts independently, but that outcomes are produced through the interplay of all these actors. They argue that animals act in the world in meaningful ways, even though their agency does not resemble human reasoning or speech. Instead, it operates through their bodies, senses and behaviors. Technologies also shape the possibilities for action, while designers and caretakers mediate interpretations of animal behavior. They consequently acknowledge the difficulty of including animal perspectives, and that this framework risks anthropomorphism or projection. Aspling and Juhlin also highlights how technology can participate in shaping asymmetrical power relations between humans and animals, meaning that researchers must remain reflexive about all these layers [2].

3 METHOD

3.1 Research Approach and Design

This thesis adopted a qualitative, abductive and iterative approach, enabling an in-depth understanding while allowing data gathering, analysis and theoretical understanding to develop alongside one another throughout the research process. ANT was used as a theoretical lens to explore how calf care emerged through relations between farmers, calves and the technological system. In this thesis, ANT was not used as a moral claim suggesting that calves, humans and the AMF

system are equal or symmetric actors. Rather, it was employed as an analytical framework that enables the inclusion of nonhuman actors in the analysis, acknowledging that the significance and influence of actors may vary across different aspects of the socio-technical system.

For this thesis, a qualitative case study of AMFs in use on Swedish dairy farms was conducted, drawing on ethnographic and contextual inquiry-inspired qualities. Since the thesis was concerned with how care was enacted through everyday routines and situated practices, it was important not only to examine what participants said about the systems, but also how care practices unfolded in context. Contextual inquiry emphasizes understanding practices within the environments where they naturally occur [21], making it especially relevant for studying socio-technical systems in HCI. To capture this situated dimension of calf care, contextual interviews conducted on-site at farms were prioritized whenever possible and relevant. This made it possible to combine interviews with direct observations of feeding systems in their everyday settings, supporting a richer understanding of how care was enacted in practice.

At the same time, interviews conducted as sit-down sessions were suited for participants whose expertise extended beyond immediate farm work. Participant selection based on variation is commonly used to gather perspectives and experiences from a broader range of people [9]. In this sense, the inclusion of both practitioners and professionals for this thesis made it possible to approach AMFs from multiple angles. Farm workers contributed understandings grounded in everyday care work and interaction with calves, while veterinarians, advisors and researchers provided broader reflections on calf health and technology use across multiple farm contexts.

Essentially, interviews were used to make it possible to access diverse participants' own experiences [13]. They followed a semi-structured format in order to support openness for reflections initiated by participant, while still ensuring that certain topics were consistently covered and allowing recurring themes to emerge. Semi-structured interviews are particularly useful when the aim is to collect open-ended qualitative data and explore participants' experiences and reflections in greater depth through flexible questioning [10].

To understand calves as active participants, Moreira's concepts of representational and material participation were applied together with Aspling and Juhlin's perspective on animal agency within socio-technical systems [2, 34]. Farmers' interpretations of calves therefore became important forms of representational participation, while calves were also approached as materially participating actors through the ways they shaped, adapted to and sometimes resisted the feeding system in practice.

3.2 Participants

A total of nine participants were included in this study. They were recruited through university contacts and by contacting relevant professionals and farms identified through online searches. Seven participants were female, while two participants were male. They ranged in age from 32 to 66 years, with three participants in their 30s, three in their 40s, one in their 50s and two in their 60s.

Three participants worked at a large farm where calves were fed using an AMF system. In addition, two participants worked at another large farm where an AMF system had previously been used but had later been replaced with manual feeding, providing insight into the challenges and reasons behind its discontinuation. At this farm, calves were bucket-fed and milk was distributed using a milk taxi. Across both mentioned farms, participants held varying responsibilities related to practical work in the barns. Their roles included barn managers, as well as positions more focused on veterinary care or the management of technology. In addition, one participant operated a smaller dairy farm without an AMF system, where calves were also bucket-fed, to provide a comparative perspective on alternative feeding practices.

Three participants held expert roles beyond farm work. This included one veterinarian with experience working with calves and conducting research on technology use in dairy farming, including AMF systems, and one advisor with experience in AMF-related research and educational work within animal welfare. In addition, one participant had extensive research experience on the subject, as well as prior experience working as both a veterinarian and advisor.

3.3 Data Gathering and Analysis

Data collection consisted of interviews in combination with some observation, where those with farm workers were conducted on-site at the farms as planned, except for the one with the participant from the smaller farm, which was conducted remotely due to geographical constraints. With the exception of one interview, all interviews were audio recorded and later transcribed. Interviews ranged in duration from approximately 45 to 90 minutes. One shorter interview, lasting approximately 20 minutes, was documented with notes instead of audio recording.

The interviews were iterative and mutually informative, meaning that insights gained from earlier interviews shaped later ones and enabled deeper exploration of emerging themes. As a result, the interview guides were different for each session. They continuously developed throughout the process and were adjusted based on what appeared most relevant in the material. For example, the stockperson's eye emerged as particularly central for understanding how care and attentiveness were structured, while questions about animal-farmer relationships became somewhat less central than initially expected. Since participants also held different roles and forms of expertise across the field, the interviews needed to be adapted accordingly. Rather than following the interview guides rigidly, interviews developed around the participants' own experiences and reflections, while still ensuring that core themes relevant to the thesis were covered.

Three of the interviews were conducted jointly with another student carrying out a separate thesis project. During these interviews, each researcher asked their own set of questions related to their respective thesis. However, material and insights that emerged from the other student's questions were also included in the analysis when considered relevant to the present thesis.

Data analysis followed a thematic approach focusing on the ANT perspective. Interview transcripts and notes were first reviewed and coded to capture meaningful segments related to participants' opinions about and experiences with calves and different feeding configurations. The coding process was inductive at first, aiming to identify more data-near interpretations, which was beneficial in keeping the codes closely connected to participants' accounts and avoiding overly abstract interpretations [7, 14]. It was only later, during the development of sub-themes, that the codes were interpreted more theoretically and therefore drew on a more deductive approach where the research question and the theoretical lens of ANT was central [4, 14]. These sub-themes were then further grouped into broader main themes through an ongoing process of refinement. Table 1 shows the development from detailed codes to sub-themes and main themes. The initial codes were kept in Swedish, as all interviews were conducted in Swedish and the coding process aimed to remain close to the empirical material at first.

Main Theme	Sub-theme	Example Code
Reorganizing Care Practices	AMFs reduce physical workload and save time	Där tungt och kämpigt att fylla på och lyfta hinkar
	Balancing automated monitoring with physical presence in everyday routines	Icke-målinriktad närvaro (gå runt och kolla läget) är viktigt

	Farm-specific conditions determine what solutions work	Hur väl man fortfarande kan behålla närvaro till kalvarna med kalvamma beror på hur stallet är uppbyggt och organiserat
	Manual feeding systems routines and integrated observation	Utfodring som morgonrutin möjliggör tidig upptäckt av förändringar efter natten
	Staff training	Tar ett tag för personal att lära sig nya system innan man får erfarenheter och vet vad som funkar bäst
	Newborn calves need hands-on care and early human contact	Nyfödda kalvar är känsliga och behöver mycket hjälp, automatisering svår
The Socio-Technical Stockperson's eye	Physical signs of calf health	Ser att en kalv mår bra på beteende, springer och leker
	Perception combining digital monitoring with physical observation	Digitala och mänskliga signaler vägs samman för att avgöra hälsotillstånd
	Remote monitoring capabilities	Kalvamma ger data om konsumtionsmönster under längre tid, vilket är hjälpsamt
	Farmer attitude and personality	Personliga egenskaper påverkar djurögat, vissa är mer uppmärksamma och engagerade än andra
	Technology as support and not replacement of care	Relation till djuren är viktig för säkerhet i arbetet
	Developing stockperson's eye through experience and practice	Erfarenhet av vad som är normalt är avgörande för att kunna upptäcka avvikelser hos kalvar
	Technology cannot fully replicate the stockperson's eye	Människor kan väga ihop flera signaler, vilket tekniken har svårt att göra
Distributing Responsibility and Control	AMFs provide overview and control	Bättre översiktlig koll med automatiska system, särskilt vid mycket djur
	Vulnerability when AMF reduces workforce	Nackdel med litet begränsat team som är upplärd med kalvamman när det blir mycket

	Staff competence and individual differences	Kan hända att personal med sämre kunskap om datorer och systemet gör fel
	Remote monitoring and mental workload	Vissa får stress av att vara uppkopplad till stallet hela tiden
	Technical failures and system reliability concerns	Teknikberoende skapar sårbarhet när system eller uppkoppling fallerar
	Trust in team	Man delar ansvar och har förtroende för teamet
	Manual control and human responsibility benefits	Direkt och omedelbar bedömning i tid och rum vid manuell utfodring
	Stress and human imperfections	Tids- och resursbrist påverkar uppmärksamhet och kan leda till att problem ignoreras
Calves as Active Participants	Individual calf behavior and personality differences	Kalvar är individer som är olika
	Calves learning AMF	Strular med vissa kalvar att lära sig AMF, men är sällan
	Calves manipulating AMF	Vissa kalvar lär sig klämma sig in i kalvamman, buffa bort andra kalvar och ta deras giva när de är små
	Social hierarchy and group dynamics among calves	Social hierarki mellan kalvar med lågrangare och högrangare
	Calf autonomy with AMF	Kalvar väljer själva när de går och dricker

Table 1. Overview of main themes, sub-themes and example codes identified through thematic analysis.

3.4 Ethics

Ethics was integral to the design of this study rather than treated as an afterthought. Several principles guided the research approach, aiming to ensure that both humans and animals were treated with care and respect.

The research process was grounded in reflexivity, acknowledging the position of both the researcher and caretakers. Particular attention was given to asymmetrical power relations between humans and animals, and to the ways these might shape observations and interpretations [2, 32, 49]. It was recognized that animals could not verbally communicate consent or participate in the same way as humans. The thesis therefore does not claim to fully have understood the animals involved in the study or accurately represented their perspectives.

The thesis was also careful not to adopt a judgmental stance toward farmers and their care practices. Instead, farmers' expertise and decisions were treated as central to the functioning of AMFs. The study was designed so that participants could take part comfortably and provide fully informed consent, with no hidden critiques embedded in the research objectives. The focus was not on criticizing individual farmers, but on understanding broader socio-technical systems. Informed consent was obtained before interviews and observations through verbal information about the purpose of the study and how the collected material would be used. All identifying information was anonymized, and data was handled

securely and not stored longer than necessary. Particular care was also taken to ensure that farm safety routines were followed. Lastly, the decision not to include more detailed descriptions of individual participants was a conscious choice aimed at reducing the risk of identification within a small and specialized field.

4 FINDINGS

Through the analysis of the interview material, four main themes were identified: *Reorganizing Care Practices*, *The Socio-Technical Stockperson's Eye*, *Distributing Responsibility* and *Control and Calves as Active Participants*.

4.1 Reorganizing Care Practices

This theme captures how the introduction of AMFs does not simply reduce or replace care work, but rather reconfigures how calf care is organized, in line with the first research question. In manual feeding systems, that is bucket feeding with or without a milk taxi, care is structured around routine tasks of preparing milk and feeding all calves the same amount at fixed times. Participants explain how these activities require direct and repeated physical engagement with calves, creating regular and natural opportunities for observation and intervention if needed. With the introduction of AMFs, these routines are partly replaced by continuous system monitoring. Feeding becomes automated and individualized, allowing calves to drink multiple times per day without direct human involvement, as explained by participants. As a result, care is no longer organized primarily around routine feeding moments, but instead shifts toward monitoring feeding patterns. From an ANT perspective, this illustrates how care is reconfigured through changing relations between human and non-human actors, as the AMF system does not simply assist existing routines, but actively transforms them.

Physically demanding and time-consuming tasks are also dramatically reduced with AMF systems according to participants. Through an ANT lens, this illustrates how technology participates in shaping what care work consists of. The findings indicate a shift in labor from primarily physical to increasingly cognitive and interpretive work. Farmers must learn to navigate and understand the system, making competence and experience with interpreting data a central aspect of care. There is therefore partly a transition from "doing" care through physical tasks to overseeing care through the system, where attention is directed toward data and alerts.

Another key distinction is how physical presence and attentiveness to calf health are organized differently in manual and automated feeding systems. Manual feeding creates regular opportunities to observe the animals, enabling an immediate and direct overview of calf health. When arriving in the morning, farmers can quickly notice if something is wrong, as observation is built into the act of feeding itself. "At the same time, you observe how the calf is doing and how it behaves … you're doing two tasks at once" (P6, barn manager at a farm that stopped using an AMF), explains one participant. In this sense, attentiveness is not something that needs to be actively decided, it is structurally integrated into the practice of care.

In contrast, with AMFs, physical presence is separated from feeding. While the system provides continuous data and monitoring, physical observation becomes less automatically integrated into daily routines. Importantly, this does not necessarily mean that farmers lose contact with the calves. Rather, the findings suggest a shift in how presence is organized. Physical presence becomes something that must be actively chosen and maintained, rather than something inherently required by the task. Farmers emphasize that it is still necessary to regularly enter the calf area and observe animals directly. However, maintaining such presence appears to depend on an active decision not to rely too heavily on system data. This introduces variation in how care is enacted, where sustaining close observation may depend on how careful or attentive a caretaker is, rather than being structurally enforced by the feeding routine.

A recurring pattern amongst participants is the importance of presence that has no specific goal, that is, moments where farmers using an AMF system "go in and look around" without a specific task or predefined purpose. "You can't just stop looking at the animals just because you have all this help" (P1, barn manager at a farm currently using an AMF) argues one participant. This form of presence is described as central to care, as it allows farmers to maintain a general sense of the calves' condition beyond what is captured through system data. Participants emphasize that this kind of observation is not directed at a single calf or triggered by a specific alert, but instead involves scanning the environment and building an overall impression of what is normal and what may stand out. The findings demonstrate that it is through this ongoing, informal engagement that subtle deviations can be detected early, often before they are registered by the system. This further shows that automation does not replace care practices, but instead shifts how attentiveness must be actively maintained by farmers.

This highlights how, in this specific aspect of the socio-technical system, human actors play a particularly important role, as sustaining close observation depends largely on how caregivers choose to engage with both calves and technological monitoring. Yet, from an ANT perspective, it also illustrates how attentiveness does not reside solely within either the farmer or the technological system, but instead emerges through relations between them. The findings suggest that the AMF reconfigures the socio-technical conditions through which attentiveness is achieved. Whereas manual feeding routinely brought caregivers into close contact with calves, the AMF weakens this connection by enabling feeding to occur without physical presence.

Manual feeding practices are furthermore often still retained and valued in specific situations, particularly in early life stages or when calves are unwell, as explained by participants from one farm. These moments are seen as offering closer contact and more direct insight into the animals. This indicates that, in more sensitive and vulnerable care situations, farmers continue to rely on physical and embodied forms of attentiveness that are difficult to support by technology. It shows how human sensitivity is still not understood as fully translatable into technological systems. Drawing on ANT, this suggests that care cannot be fully delegated to technological actors, as certain forms of attentiveness continue to emerge through direct bodily engagement between farmers and calves.

This theme further suggests that the reorganization of care through AMF systems is highly contextual and dependent on specific farm conditions. Rather than producing a uniform way of working, AMFs are integrated differently across farms. They appear to function more effectively when a smaller, consistent group of workers is responsible for the calves and familiar with the technology. In contrast, environments with high staff turnover or many different caretakers may lead to less consistent monitoring and weaker system understanding, as told by one participant. The findings additionally suggest that the physical placement of calves, feeders and system interfaces influences how easily farmers can combine digital monitoring with physical observation. For example, when systems are located in ways that encourage proximity to the calves, it becomes more natural to integrate regular physical presence into daily routines, according to one participant. The time available for observation is also shaped by workload. Participants explain that when they have many competing tasks, there is a greater risk that attention shifts toward system data rather than direct animal observation. These factors highlight that AMFs do not determine care practices on their own. Instead, care emerges through situated configurations in which technology, staffing and farm structures interact.

4.2 The Socio-Technical Stockperson's eye

One of the most central themes that emerged from the findings is the importance of what participants referred to as the *stockperson's eye*. This concept appeared repeatedly across interviews, often introduced by participants themselves without being prompted, indicating its strong relevance in everyday calf care practices. The stockperson's eye can be understood

as a key way in which care is enacted, as it relates to the ability to notice, interpret and respond to subtle changes in calf health and behavior. One participant described it as follows:

> "Some people, through spending time with animals, develop a feel for them and are able to see better when something is wrong. It is not always something that comes naturally, and it may not be something that everyone is equally good at" (P4, veterinarian and PhD student).

While several participants described the stockperson's eye as something partly innate, it was also consistently framed as a skill that develops over time. It is shaped through experience and repeated exposure, where a person eventually learns to recognize patterns in animal behavior. At the same time, participants explain that it is not only a matter of skill or training, but also of individual attitude and engagement, suggesting that attentiveness is influenced by how much care and attention a person is willing to invest. The findings also demonstrate that the stockperson's eye is not a fixed ability, but rather something that is continuously developed. Some participants described how their ability to detect early signs of illness in calves improved over time, as they became more familiar with what is considered "normal" behavior. They explain that this process involves learning to recognize subtle deviations, such as changes in appetite, posture, movement or general demeanor. Importantly, this form of knowledge is largely experiential, rather than easily articulated. Participants often described it as something they "just see" or "feel". This suggests that attentiveness in calf care relies heavily on tacit and embodied knowledge that is difficult to fully formalize within technological systems. It also demonstrates, through an ANT perspective, that the stockperson's eye cannot be understood as a fixed or isolated human ability, but rather as something that continuously emerges and develops through ongoing relations with calves.

This also means that the stockperson's eye is shaped by work practices and routines. Feeding systems that enable frequent and close interaction with calves may support the development of this ability, while systems that reduce direct contact may change how it is enacted. In some interviews, participants raised concerns that the introduction of AMFs could risk weakening the stockperson's eye, particularly if farmers rely too heavily on system data and reduce their physical presence among the calves. These findings demonstrate a concern that attentiveness may become more distant or mediated, potentially limiting opportunities to develop or maintain this form of embodied knowledge. This reflects similar concerns found in previous PLF research regarding technologically mediated distance between humans and animals.

However, across most interviews, a different pattern emerged. Rather than disappearing, the stockperson's eye is more often described as being reconfigured or even strengthened through the integration of AMFs. Participants describe how data on consumption patterns can highlight deviations in behavior over time, which might otherwise be difficult to detect, especially in larger groups of calves. In this sense, technology can support and extend perception and attentiveness. At the same time, participants consistently emphasize that technology alone is not sufficient. While systems can indicate that something may be wrong, they cannot interpret the full context or take action, describes one participant. This highlights that the stockperson's eye remains dependent on human judgment and abilities. Another participant noted:

> "Farmers often tell me that technology cannot replace the human stockperson's eye. It just isn't possible, because technology will always produce false positives or false alarms in one way or another. You could say that the alerts extend their presence among the animals. When they receive an alert, they go there, and the human stockperson's eye meets the animal, so to speak, through that signal" (P8, advisor and animal welfare educator).

Through an ANT lens, the stockperson's eye becomes a socio-technical form of attentiveness, where technological systems participate in shaping what becomes visible and actionable, while human interpretation remains necessary for contextualizing and responding to these signals.

Furthermore, the findings suggest that the ability to interpret system data correctly is itself grounded in prior experience with calves. Participants described that without an understanding of what constitutes normal and abnormal behavior, data risks being misinterpreted or lacking meaning. They therefore stress the importance of combining digital information with physical observation in order to form a more complete assessment. This suggests that data may not always capture the full situation, and important signs can still be missed if farmers rely too much on system outputs. When these different forms of knowledge are combined, that is system data, physical observation and experiential understanding, a hybrid *socio-technical stockperson's eye* can emerge.

From an ANT perspective, attentiveness emerges through relations between multiple actors rather than residing solely within the human caretaker. Perception is not diminished but expanded, as farmers are able to draw on multiple sources of information to identify and respond to calf needs. Rather than replacing embodied knowledge, technology becomes integrated into it, resulting in a more comprehensive form of care.

4.3 Distributing Responsibility and Control

This theme highlights how AMFs reshape the experience of control and responsibility in calf care. Participants describe how the system provides an increased sense of overview, as farmers can monitor calves through system data and access this information remotely via mobile phones. These findings suggest a form of centralized and continuous control, where farmers can check the system at any time, even when they are not physically present in the barn. Participants describe this as providing reassurance, as it becomes easier to keep track of calves and detect potential issues. This illustrates how digital connectivity reconfigures the temporal boundaries of care work by extending attentiveness beyond physical presence on the farm.

However, this increased control is also tied to a sense of continuous responsibility. The system is always accessible, even outside working hours. The findings suggest that this creates a situation where the boundary between work and non-work becomes less clearly defined. Some participants describe difficulty switching off, as the possibility of checking the system always remains present. At the same time, this experience is not uniform. While some participants experience this constant connectivity as stressful, another participant describes it as less problematic and feels able to disengage from the system when needed. This suggests that the impact of continuous monitoring depends on individual coping strategies. Importantly, this sense of ongoing responsibility is not unique to AMFs. Even in manual feeding systems, one participant describes difficulty disconnecting from their work and often continue to think about the animals outside of working hours. However, if we look at the roll of the technological systems through an ANT perspective, AMFs extend this dynamic by making the farm digitally accessible at all times, even outside of work. A key factor in managing this extended responsibility is trust in others. Participants emphasize that being able to rely on colleagues allows them to step away from constant monitoring. This demonstrates that when responsibility is distributed within a trusted team, the need to continuously check the system and the mental burden is reduced.

While AMFs can increase control, they also introduce new forms of vulnerability related to technological dependence. Several participants highlight concerns about relying on care practices that depend on stable infrastructure, such as internet connectivity. Participants therefore emphasize the importance of maintaining the knowledge and capacity to manage calves without relying on technology, in order to handle situations where the system is unavailable. Several participants also describe how the introduction of AMFs allows for fewer workers to manage larger groups of calves. While this increases

efficiency, it can also reduce the care system's capacity to respond to unexpected events. Participants explain that situations such as technical failures or sudden farm events that increase workload may become more difficult to manage with limited workers. Some express specific concerns about replacing human labor with technology, while also acknowledging that AMFs are a significant investment and farms must remain financially viable, making reduced labor requirements part of the rationale for adopting the system. Furthermore, this vulnerability is not experienced equally across farms. Some participants describe how they reduce risk by ensuring access to additional labor when needed, for example by being able to call in extra staff during peak periods. This highlights that the resilience of automated systems depends not only on the technology itself, but on how it is embedded within broader organizational structures. From an ANT perspective, resilience emerges from the configuration of the entire actor-network. Whether a technical failure becomes a serious problem depends on the availability of workers, the skills of caregivers, organizational arrangements and infrastructure.

This theme also raises the question of how responsibility is organized and distributed in systems with AMFs. In manual feeding systems, responsibility is often more direct, immediate and concentrated in specific individuals. According to participants, feeding takes place at defined moments, where a person is physically present and responsible for preparing and delivering the milk. As one participant described, this creates a sense of control and accountability: "It is also safer when you think about the amount and quality of the milk, because you always have someone who checks it. You always have a person at that moment who makes sure the calves get the right temperature and the right amount" (P6, barn manager at a farm that stopped using an AMF). In this sense, responsibility is tied to specific actions in time and space, where observation, decision-making and intervention occur simultaneously. Drawing on ANT, this illustrates a different configuration of responsibility, where care is enacted through direct bodily engagement rather than continuous technological mediation. However, participants also acknowledge that manual systems are not without error. Even experienced farmers can make mistakes, particularly under conditions of stress or when managing multiple competing tasks, describes several participants. This suggests that responsibility, while more concentrated, is still subject to human limitations.

Rather than asking whether manual or automated systems are better or worse, the findings suggest that responsibility is configured in different ways. With the introduction of AMFs, responsibility becomes more distributed across multiple actors. It is shared between multiple employees and technological components such as sensors, feeders, cleaning functions and digital interfaces. These elements work together to guide action and maintain care, but they are also potential sources of error. Responsibility therefore becomes less direct and more relational, as it depends on how different parts of the system function together. At the same time, the role of the human does not disappear, but is transformed. Farmers are responsible not only for performing care tasks, but also for interpreting system data and managing technical issues. This introduces a different kind of responsibility, which is less tied to immediate physical action and more to ongoing monitoring. Furthermore, responsibility is not only distributed between humans and technologies, but also shaped by organizational and environmental conditions. Factors such as staffing and the spatial arrangement of the system influence how effectively care can be carried out.

4.4 Calves as Active Participants

This theme highlights how calves are not passive recipients of care, but active participants who shape how AMFs function in practice. Calves are individuals whose distinct behaviors affect feeding practices, including in manual feeding systems. However, with AMFs, their participation becomes mediated through technology, as calves interact not only with humans but also with the feeder itself. This creates new forms of engagement, where calves adapt to and negotiate with the technological system.

In systems with AMFs, participants explain that calves are given the freedom to decide when to drink. Instead of being fed at fixed times, they can access milk throughout the day, which one participant describe as more aligned with natural feeding behavior. This flexibility also reflects how AMFs accommodate individual variation. According to participants, each calf is assigned an individual feeding curve and milk allowance, allowing those with greater capacity to consume more. Drawing on ANT, this highlights that the AMF does not operate independently of specific animal needs, but is partly shaped around the individuality of calves.

At the same time, calves are required to adapt to the system through caretakers teaching them how to use it. While some individuals may be extra challenging or never fully adapt, participants generally described these cases as relatively rare, with most calves adapting fairly easily. Still, calves must learn how to use the feeder themselves, and participants described how this learning process varies between individuals. Some calves learn quickly, while others require a bit more time and guidance. This highlights that calves' adaptation to the system's routines and constraints also shapes how smoothly the system operates.

Beyond learning, calves also shape how the system functions through their behavioral responses to it. Several participants describe how calves can outsmart or manipulate the feeder, for example by drinking milk intended for other calves. "When they are small, they can squeeze two into one station. So we've seen that some clever ones learn how to kind of steal food from another calf" (P2, product manager at a farm currently using an AMF system). These behaviors show that calves actively affect feeding outcomes, rather than simply following system rules. In some cases, calves also respond to system limitations. One example mentioned by multiple participants was calves biting and damaging the teat when milk was no longer available. These actions can be understood as forms of resistance or reaction to the system, further demonstrating, through the ANT perspective, that calves are active participants.

In manual systems, younger calves are often kept in individual pens, while, according to participants, AMFs are associated with earlier group housing in which feeding takes place in shared environments that enable calves to interact with one another. Similarly to earlier research, participants in this study note that calves behave differently in these group contexts and that social dynamics become part of everyday feeding practices. For example, calves form social hierarchies through subtle interactions, where some individuals display more assertiveness than others, as explained by participants. Another participant notes that subordinate calves are not necessarily isolated, but often form their own groups. They explain that these dynamics persist into adulthood and influence how barns and feeding systems must be designed, as access to resources depends on how well the environment accommodates both dominant and subordinate individuals. Another participant describes that calves differ in how assertive they are when it comes to feeding, where some individuals are more food motivated or better able to take space at the feeder, while others may be less inclined to compete.

> "I think that automatic feeders can, in many ways, give calves greater opportunities control their feeding more individually … but because there is also competition within the system, these aspects can work against each other. So if a calf is weaker, it may have less of an opportunity to do what it wants and eat in its own way" (P9, researcher).

Essentially, these differences shape how calves are able to use the system and benefit from its flexibility. From an ANT perspective, this illustrates how feeding outcomes emerge through social relations between calves that shape access to resources within the environment.

4.5 Synthesis of Findings

Together, these themes answer the first research question by showing how calf care emerges through interactions between farmers, calves and AMFs rather than isolated human action, including how calves actively participate in shaping how the systems function in practice. They also answer the second research question by demonstrating how attentiveness and responsibility become reorganized compared to manual feeding practices. While attentiveness in manual systems is structurally embedded within direct feeding routines, feeding and observation become more separated with AMFs, meaning that physical presence has to be more actively maintained by farmers. When such presence is maintained, attentiveness is enacted through a socio-technical stockperson's eye. Responsibility is similarly reorganized. Whereas responsibility in manual systems is concentrated around direct feeding moments in time and space, AMFs distribute responsibility across ongoing remote monitoring practices and a wider network of actors. While this increases the sense of continuous control, it also creates new vulnerabilities linked to infrastructure and staffing arrangements.

5 DISCUSSION

5.1 Situated Care

This thesis suggests that agricultural HCI requires a shift from optimizing systems to understanding how they are enacted in variable contexts. The findings show that AMFs do not produce a uniform way of enacting care, but are instead integrated into farm practices in highly situated and context-dependent ways. With AMFs, physical presence with calves becomes an active choice that needs to function smoothly within the interplay of system layout, staffing and workload. Agricultural technology design cannot be one-size-fits-all, and should support adaptation across farm contexts rather than standardization. From a design perspective, this points to the need for systems that can adapt to diverse working conditions, rather than enforce predefined workflows.

The findings also suggest that AMFs cannot be fully understood or evaluated outside of the specific contexts in which they are used. This resonates with HCI research emphasizing that agricultural technologies should be understood as part of broader socio-technical systems shaped by local practices. It also connects to the idea that agricultural systems only become meaningful when integrated with farmers' experiential and situated knowledge [29, 30]. In this sense, the findings further support and extend previous research by showing that similar socio-technical dynamics identified in studies of other PLF technologies are also present in the context of AMFs.

Furthermore, previous research warned that PLF technologies could risk creating greater distance between humans and animals, reducing attentiveness and transforming animals into quantifiable data points [5, 24, 38]. While such risks were also visible in this study, the findings suggest that these outcomes should not be understood as automatic consequences of AMFs themselves. Rather, the effects of AMFs depended on how the systems were integrated into everyday care practices and what role farmers allowed the technology to take within calf care. Participants described concerns that overreliance on system data could reduce physical observation and weaken opportunities to maintain a close understanding of the calves. However, whether the technology contributed to distance or attentiveness depended on contextual factors. In particular, the findings suggest an important difference between using AMFs primarily to replace human labor and reduce physical involvement, versus using them to support and expand human attentiveness. Participants who emphasized continued physical presence among the calves often described the technology as a complement to rather than a replacement for direct observation. This suggests that technologies do not determine care practices on their own, but instead shape care differently depending on how they are enacted in local farm contexts.

5.2 Integrating Technology with Experiential Understanding

The thesis also calls for designing for a new kind of perception, considering the stockperson's eye as a hybrid between data and physical perception. The findings suggest that AMFs cannot replace the stockperson's eye, but rather transform it into a socio-technical form of perception in which embodied experience and data-driven insights are combined. While system data can highlight individual deviations in feeding patterns among group-housed calves and in some ways extend farmers' awareness beyond what is immediately visible, it remains dependent on human interpretation. In practice, the socio-technical stockperson's eye occurs through an ongoing movement between digital and physical forms of observation, which are interpreted through the embodied and experiential understanding farmers develop through prolonged engagement with calves. For example, the system may indicate a change in a calf's feeding pattern, but farmers still need to physically assess the calf and draw on their sense of what is normal for that individual in order to determine whether the deviation signals illness. Previous research often describe AMFs as reducing physical labor [39, 43, 44, 46], yet the findings of this thesis show that they also generate new forms of interpretive work. Farmers were responsible for translating digital data into situated understandings of calf well-being, suggesting that automation changes the nature of care responsibilities rather than simply reducing them. This suggests that PLF technology should not be designed to replace working staff or replicate the stockperson's eye, but rather to support and remain accountable to it. Systems should support the expansion of human judgement through digital data, for example by encouraging continued physical presence and helping farmers interpret information. In this sense, effective design is about enabling meaningful integration between technology and the farmer's experiential understanding of animals.

These findings extend previous research on automatic milking systems which show that robot-generated data cannot directly determine animal health or well-being on its own, but instead requires interpretation through prior experience and continued physical contact with animals. Their findings suggest that automation may actually require a stronger stockperson's eye, since farmers need to both interpret system data and continue paying attention to behavioral and physical signs that technology cannot fully capture [29]. This closely resembles the socio-technical form of perception identified in this thesis with AMFs, where care increasingly depends on the ability to move between system data and direct observation.

This form of perception also resonates with Cort and Lindblom's concepts of remote situatedness and enhanced seeing in railway control rooms. Similar to how train traffic controllers learn to interpret abstract digital representations through embodied route familiarity and professional experience [6], farmers in this study developed a socio-technical form of perception through the integration of digital monitoring and experiential animal knowledge. Rather than simply observing data outputs, farmers learn to "see through" the AMF system, interpreting feeding patterns and alerts in relation to embodied understandings of calf behavior and health. Although farmers are not physically separated from calves in the same way that railway controllers are separated from the railway environment, AMFs introduced new forms of mediated attentiveness through digital monitoring.

5.3 Theoretical Implications

While the concepts of enhanced seeing and remote situatedness help explain how technologically mediated perception is enacted [6], ANT further contributed by highlighting how this perception emerges relationally through interactions between multiple human and non-human actors. This perspective created a different understanding of care within AMFs. Rather than viewing care as something humans perform with the assistance of technology, care instead emerged as something continuously produced through socio-technical relations. Technologies did not simply support care practices, but actively participated in shaping what became visible, how responsibility was distributed and how farmers remained attentive to the animals. Similarly, calves did not appear as passive recipients of care, but as active participants that

influenced how the system operated. In this sense, the ANT perspective contributed to moving beyond human-centered understandings of PLF technology. Importantly, while ANT is often associated with symmetry between human and non-human actors, this thesis found it useful for tracing how different actors contributed to care practices without assuming that they played identical roles within the socio-technical system.

Furthermore, the findings contribute to a One Welfare perspective by showing how animal welfare and farmer well-being become closely interconnected through everyday care practices. Participants described how stress and workload affected both their own work experiences and the quality of calf care. In this sense, the findings illustrate how human well-being and animal welfare emerge together rather than as separate concerns. At the same time, the thesis further suggests that technological systems play an important role in shaping these relations. For example, the findings showed that farmers' ability to disconnect from work was shaped through the same socio-technical arrangements that also influenced calf welfare. Rather than appearing as separate concerns, animal welfare, working conditions and technological infrastructures emerged as deeply entangled within the everyday organization of care.

5.4 Limitations and Future Research

At the same time, ANT also presents limitations as a theoretical lens. Its assumption of symmetry between actors may risk obscuring important ethical differences between humans, animals and technologies, as they do not hold the same forms of value or accountability. For example, if an AMF system fails to detect signs of illness early enough, ANT makes it possible to trace how this outcome emerged through relations between multiple human and non-human actors. However, technology itself cannot be held morally accountable for mistakes and calves cannot be understood as morally responsible actors. Responsibility for animal welfare still ultimately rests with humans.

Similarly, although this thesis seeks to include calves as active participants within socio-technical systems, it is important to recognize that humans and animals do not participate under equal conditions. Farmers retain authority, while calves remain dependent on systems of care managed by humans. Furthermore, calves' participation can never be fully accessed or represented in the same way as human experiences. Their responses can only be interpreted indirectly through human observations of behavior. While ANT was valuable for analyzing how care practices emerged relationally across different species, it is less equipped to address ethical differences between them, as well as questions of power and responsibility toward animals. This highlights both the possibilities and limitations of ANT within agricultural HCI research.

It is also important to acknowledge how the researcher's disciplinary background influenced the interpretations presented in this thesis. Livestock farming is a highly practical, experience-based and context-dependent field, where much knowledge is embodied, tacit and difficult to fully articulate to outsiders, something that was also explicitly reflected upon by one of the participants. Coming from an HCI background rather than an agricultural one, the interpretations in this thesis should therefore be understood as one attempt to connect more abstract theoretical perspectives from a different field to a highly practical farming context. While this interdisciplinary perspective made it possible to identify socio-technical dimensions that may otherwise remain less visible, it may also create tensions or gaps between theoretical language and lived farming practices. Further interdisciplinary and participatory research involving closer collaboration between HCI researchers, agricultural researchers and practitioners would therefore be valuable for continuing to explore these intersections.

Another limitation that should be noted is that the study was based on a small qualitative sample and especially focused on a limited number of dairy farms. The findings should therefore not be understood as representative of calf care or AMF use, but rather as a situated case study that revealed patterns and perspectives that also resonate with similar research on

other agricultural technology. These findings would therefore be interesting for future research to explore further, both in relation to AMFs and other agricultural contexts. Future research could also explore more direct animal-centered approaches for investigating how livestock animals engage with technological systems. In general, there is a need for more ACI research focusing on PLF technologies, as well as studies that explore entanglement theories and more-than-human perspectives on animal care, especially considering a future towards a One Welfare approach where animal welfare, human well-being and environmental considerations are understood as deeply interconnected.

6 CONCLUSION

In conclusion, this thesis has shown that AMFs do not simply automate calf feeding, but actively participate in reconfiguring how care is organized and enacted on dairy farms. Through an ANT perspective, care emerged as relational and socio-technical, shaped through interactions between farmers, calves and the technological system. The findings highlight that attentiveness and responsibility are not replaced by automation, but transformed through new forms of monitoring. At the same time, the study emphasizes that technologies cannot be understood separately from the situated practices and contexts in which they are used. By contributing with a more-than-human perspective on livestock technology, this thesis highlights the importance of understanding agricultural technologies not only in terms of efficiency and productivity, but also in relation to care, animal welfare and human well-being.

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